Assignment 03

1. Global methane levels from 2002

1.1

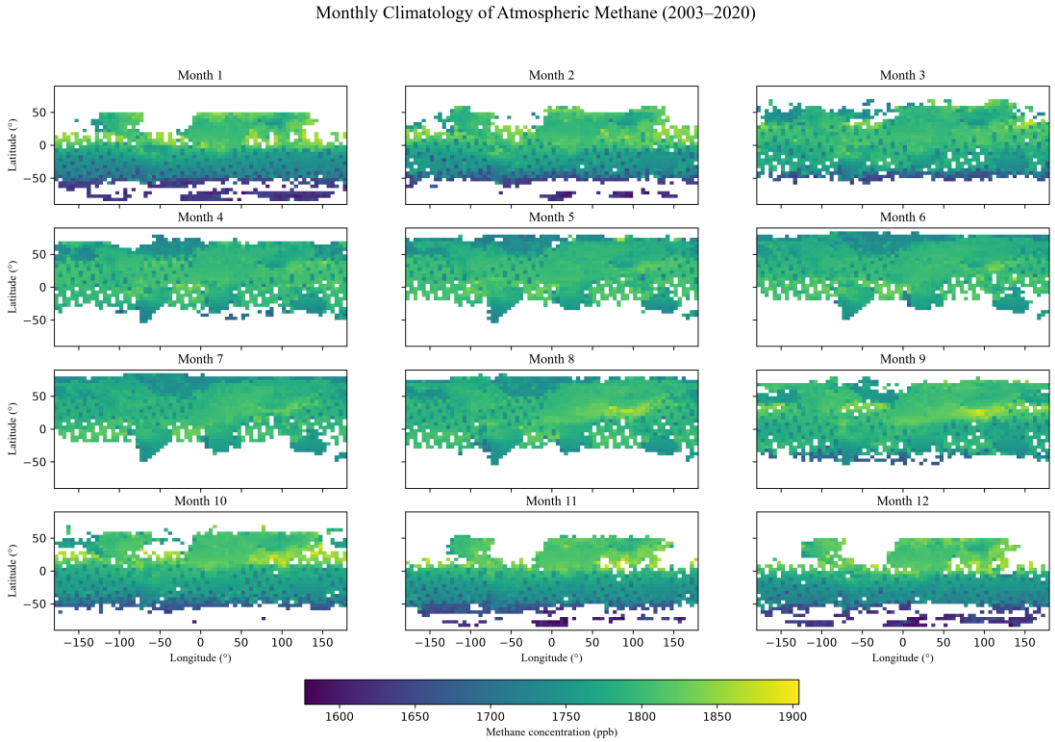


Figure 1 Monthly Climatology of Atmospheric Methane (2003 - 2020)

1.2

Figure 2 shows that the methane concentration exhibits a clear long-term upward trend, increasing from approximately 1705 - 1710 ppb in early 2003 to nearly 1,850 ppb by mid-2020, representing a net increase of approximately 140 - 145 ppb. Superimposed on this overall trend are distinct seasonal fluctuations, with an annual cycle and an amplitude of approximately 10 - 20 ppb. Temporally, methane concentrations increased relatively slowly between 2003 and 2006, with a noticeable decline around 2004. From approximately 2007 onward, concentrations resumed a sustained increase, which became more pronounced after 2014. Between 2019 and 2020, the upward trend continued, with concentrations approaching the highest levels observed during the study period.

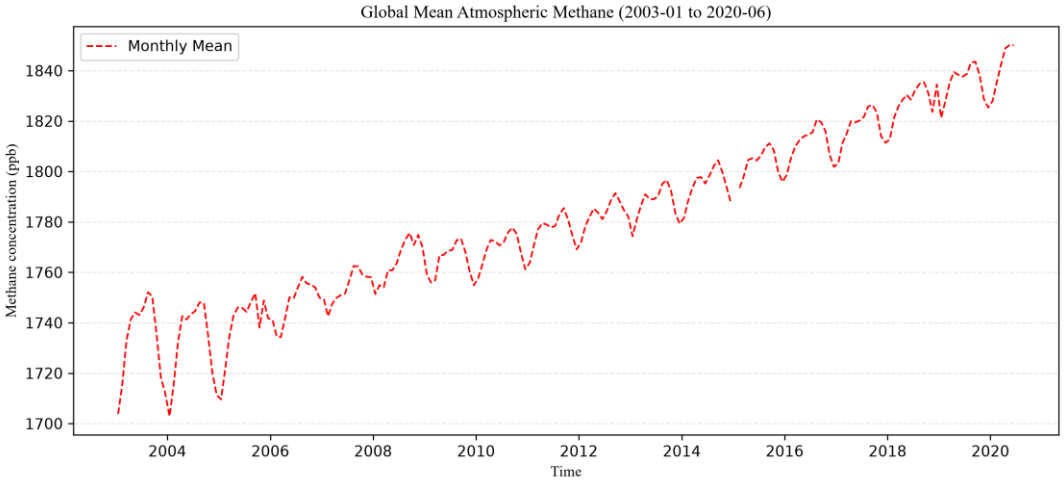


Figure 2 Global Mean Atmospheric Mrthane (2003 - 01 to 2020 - 06)

1.3

The methane anomaly time series exhibits distinct temporal variations between 2003 and 2020 (Figure 3). During the early period (2003 - 2006), negative anomalies predominated, with several substantial deviations ranging from approximately -40 to - 80 ppb, indicating that methane concentrations were considerably below the long-term mean. Between 2007 and 2013, anomalies generally remained close to zero or slightly negative, with relatively limited fluctuations, corresponding to a period of slower global methane growth. From approximately 2014 onward, positive anomalies became increasingly dominant, with both their magnitude and frequency increasing. After 2016, positive anomalies occurred more frequently, reaching a maximum of nearly +60 ppb, with particularly pronounced positive deviations during 2018 - 2020.

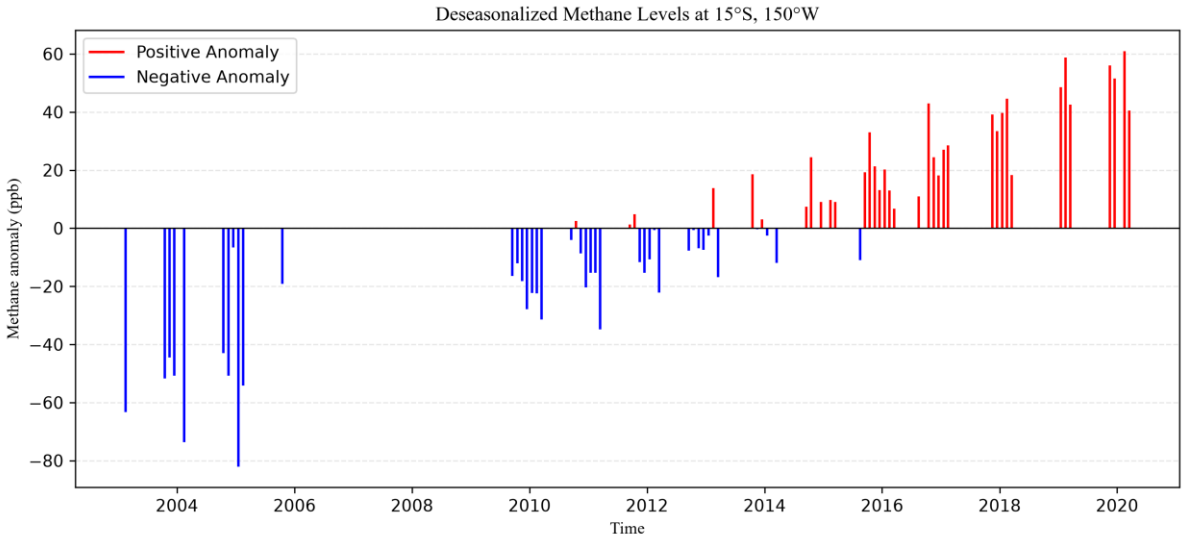


Figure 3 Deseasonalized Methane Levels at 15°S, 150°W

2. Niño 3.4 index

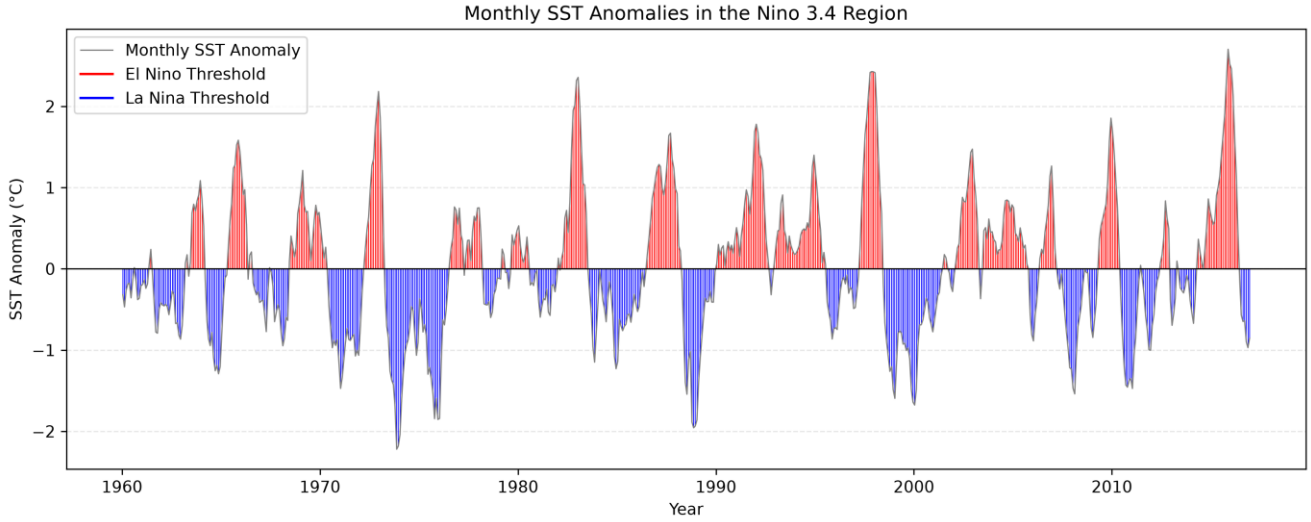


Figure 4 Monthly SST anomalies in the Niño 3.4 Region

3. Explore a netCDF dataset

This analysis uses the GPCP (Global Precipitation Climatology Project) Global Monthly Precipitation Dataset to investigate global precipitation characteristics from five perspectives: time series of global monthly precipitation anomalies, global average precipitation, longitude variation of monthly average global precipitation, spatial distribution of global multi-year average precipitation, and spatial distribution of global monthly average precipitation.

3.1

Figure 5 illustrates the interannual variations in global precipitation anomalies from 2000 to 2025. Positive anomalies indicate precipitation above the climatological mean, whereas negative anomalies represent below-average conditions. The time series exhibits considerable variability, with predominantly negative anomalies during 2008 - 2012, followed by a relatively stable period from 2013 to 2015. A pronounced positive anomaly occurred around 2016, exceeding 0.25 mm/day. After alternating fluctuations during 2017 - 2022, positive anomalies became more persistent and intensified during 2023 - 2025. Overall, the results highlight substantial interannual variability in global precipitation, with distinct periods of relatively wet and dry conditions.

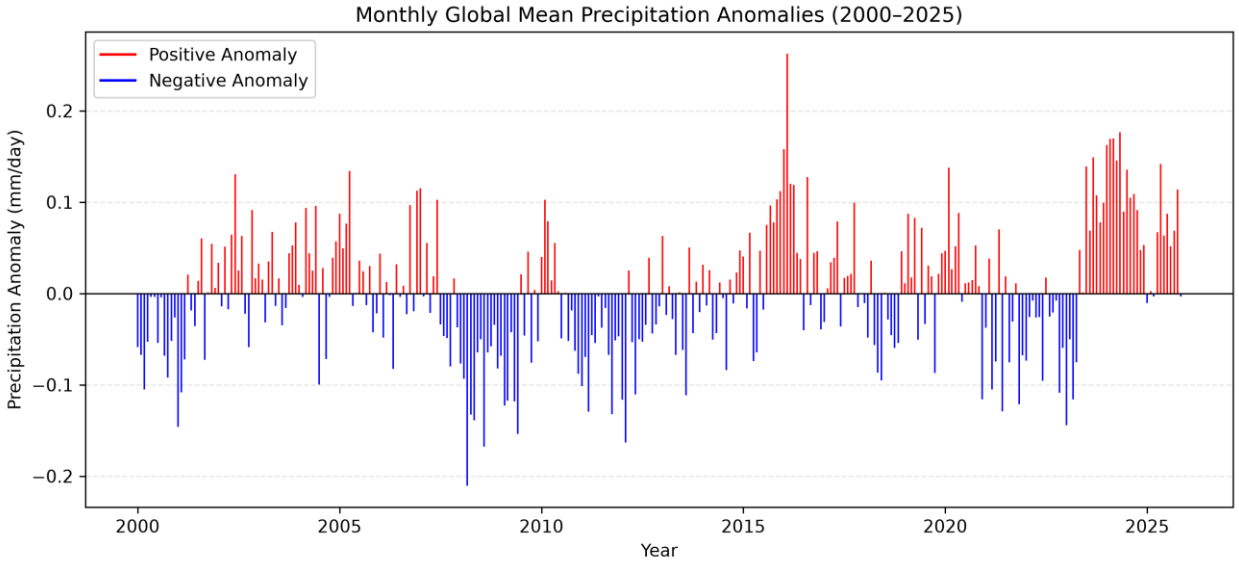


Figure 5 Monthly Global Mean Precipitation Anomalies (2000 - 2025)

3.2

3.2.1 Monthly Climatology of Global Mean Precipitation

Figure 6 shows that global monthly mean precipitation exhibits relatively modest seasonal variations, ranging from 2.96 to 3.16 mm/day, with an amplitude of approximately 0.19 mm/day. Precipitation reaches its annual minimum in April, followed by a marked increase from May to July, peaking at 3.125 mm/day in July. It then gradually declines through October before increasing again toward December, reaching 3.10 mm/day. Overall, the seasonal pattern is characterized by a summer peak, a spring minimum, and a secondary increase toward the end of the year.

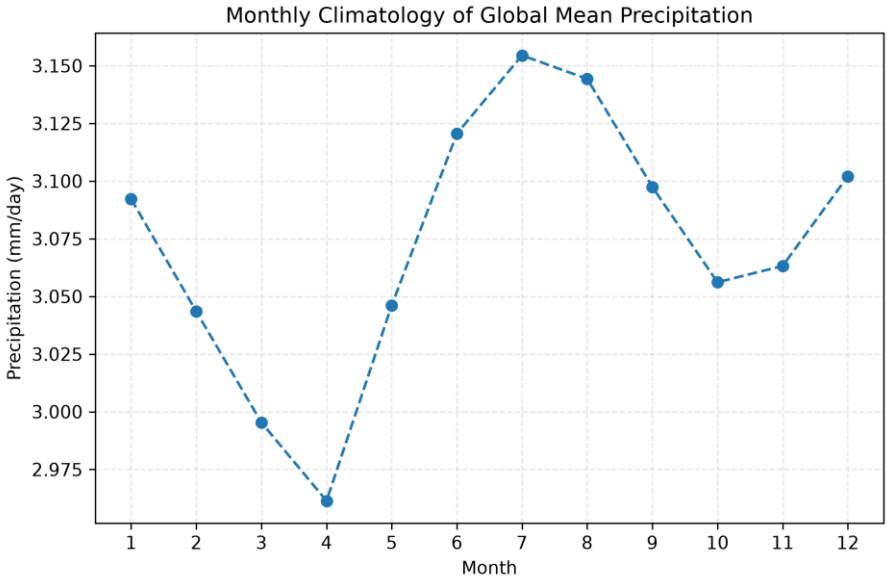


Figure 6 Monthly Climatology of Global Mean Precipitation

3.2.2 Global Mean Precipitation

Figure 7 illustrates the spatial distribution of global mean precipitation from 2000 to 2025. Precipitation exhibits a pronounced latitudinal pattern, with the highest values concentrated in tropical regions. Particularly high precipitation occurs along the Intertropical Convergence Zone (ITCZ), especially over the western Pacific warm pool, the Maritime Continent, and parts of the Amazon Basin, where local values exceed 20 mm/day. Other tropical regions, including Central Africa and Southeast Asia, also experience relatively high precipitation.

In contrast, precipitation is generally lower in the subtropical regions around 20° - 30° north and south, particularly over major arid areas such as the Sahara and Australian deserts. The polar regions also exhibit relatively low precipitation compared with the humid tropics.

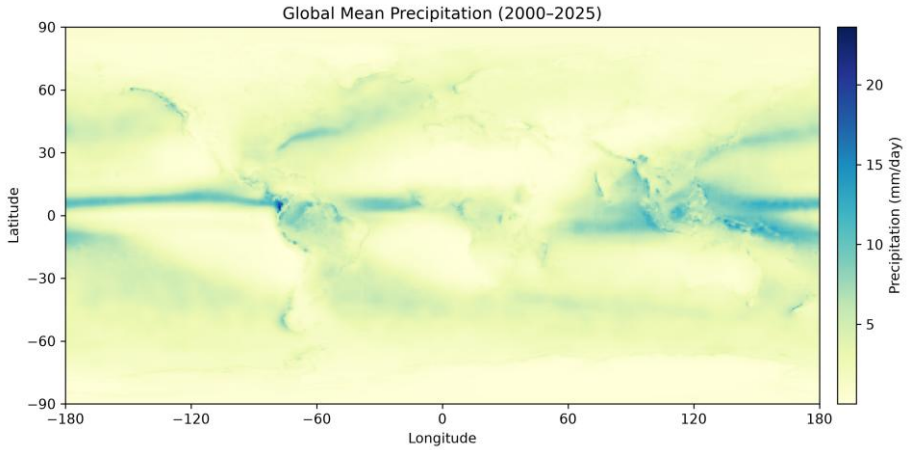


Figure 7 Global Mean Precipitation (2000 - 2025)

3.2.3 Monthly Precipitation Climatology by Latitude

The monthly precipitation climatology (Figure 8) reveals a clear seasonal migration of global precipitation belts. The main tropical precipitation belt shifts northward from approximately 0° - 5°N during January - April to 5° - 10°N during May - September, reaching its greatest intensity in July - September, with local precipitation exceeding 8 mm/day.

Relatively high precipitation also occurs between 30°S and 45°S, particularly from May to September, and around 35° - 40°N during November - February. In contrast, precipitation remains relatively low in the subtropical regions (20° - 30° in both hemispheres) and at high latitudes above 70°.

Overall, the figure highlights the seasonal migration of global precipitation belts, particularly the northward and southward movement of the ITCZ.

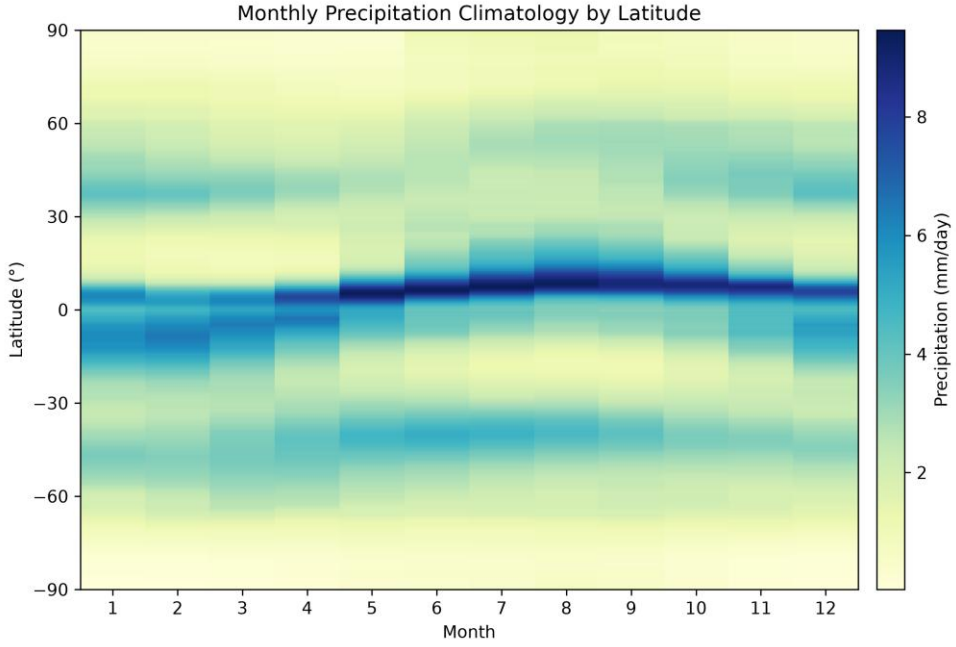


Figure 8 Monthly Precipitation Climatology by Latitude

3.2.4 Annual Global Mean Precipitation

Figure 9 illustrates the interannual variations in global annual mean precipitation from 2000 to 2024. Overall, precipitation exhibits considerable year-to-year fluctuations, with a long-term mean of approximately 3.07-3.08 mm/day. Following relatively stable conditions during 2000-2007, precipitation declined sharply in 2008 and subsequently recovered. A notable increase occurred during 2014-2016, followed by a period of lower and more variable precipitation between 2017 and 2022. From 2023 onward, precipitation increased markedly, reaching its highest level in 2024.

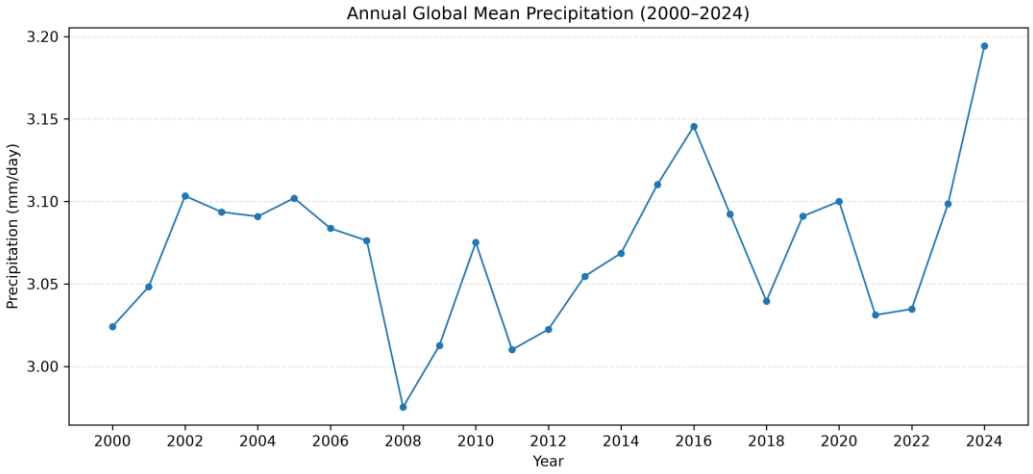


Figure 9 Annual Global Mean Precipitation (2000-2024)

3.2.5 Monthly Climatology of Global Precipitation

The monthly precipitation climatology reveals pronounced seasonal variations in global precipitation patterns. During January–March, precipitation is mainly concentrated in the tropics, particularly over the western Pacific and Maritime Continent. From April onward, the tropical precipitation belt shifts northward, with enhanced precipitation over South and Southeast Asia during June–August, reflecting the influence of the Asian summer monsoon. During October–December, the precipitation belt gradually shifts southward, with increased precipitation across parts of the Southern Hemisphere tropics. The Amazon Basin and Central Africa maintain relatively high precipitation throughout the year, despite seasonal variations in intensity and extent. Overall, the figure highlights the seasonal migration of tropical precipitation belts, driven largely by the movement of the ITCZ and regional monsoon circulation.

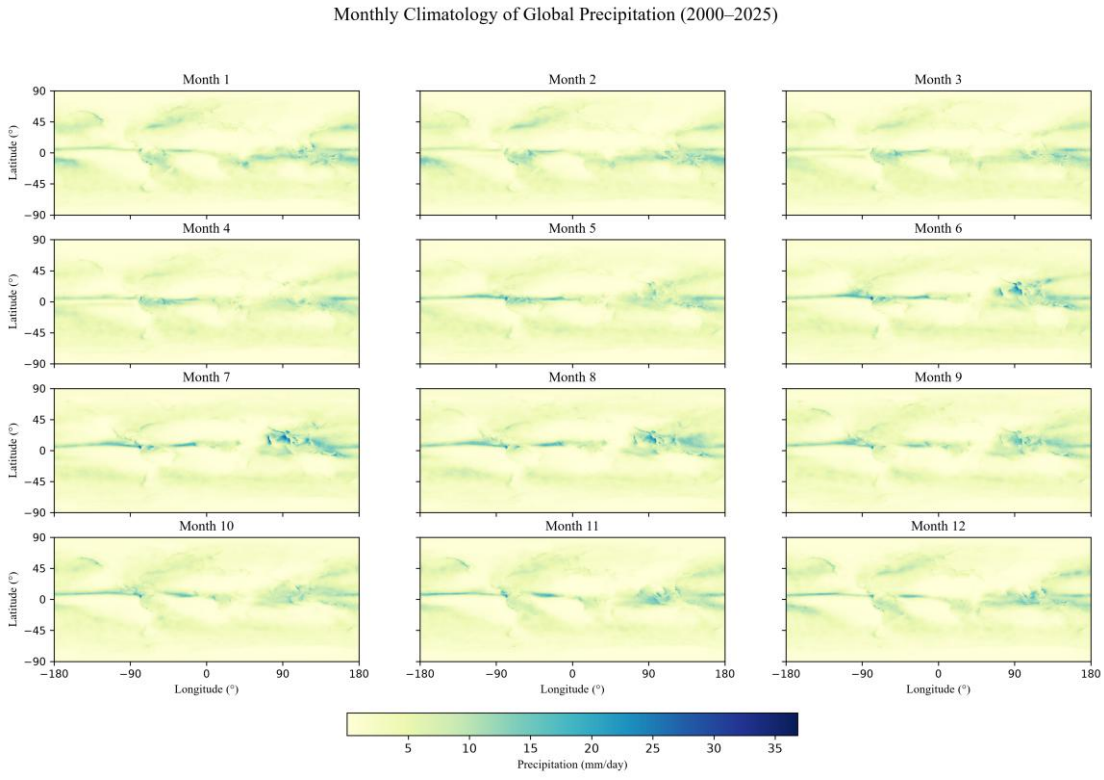


Figure 10 Monthly Climatology of Global Precipitation (2000-2025)